



Merchant Transactions Data Analysis

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Executive Summary

Objective

Analyze transaction data to understand merchant performance, trends, and key drivers of TPV.

Methodology

- **SQL:** Data cleaning, ID mapping, and cohort dataset creation.
- **Tableau:** Dashboards for tenure, MTD amounts, onboarding cohort trends, and top merchant TPV trends
- **Python:** Time-series analysis focusing on seasonality and concentration.

Key Findings

- Volume is highly concentrated among a small number of merchants.
- Aggregate trends are volatile and largely driven by outliers.
- No strong or consistent seasonality observed across time-based analysis.

Key Takeaway

A small subset of merchants disproportionately influences overall performance, limiting the reliability of aggregate trends.

Dataset Overview

Total Rows

262,771

Time Period

Aug 2020 - Jun 2021

Distinct Merchants

19

Scope & Mapping

- Transaction-level data across multiple merchants, countries & currencies.
- Merchant ID mapping provided (old - new IDs).

Observations

- Merchant onboarding concentrated within a few months.
- Minor data inconsistencies (e.g., transactions before registration).

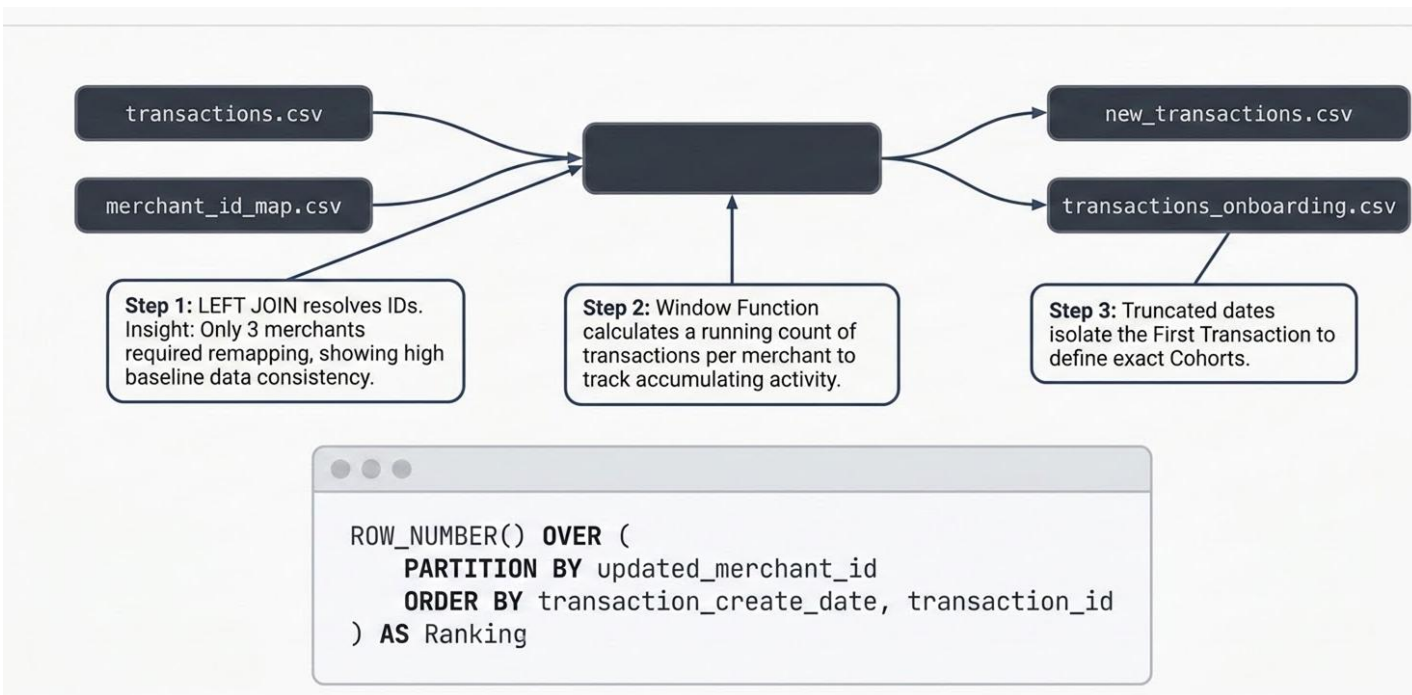
Insight

Relatively small Merchant Base - sensitive to individual merchant behavior

SQL - Data Prep & Analytical Foundation

Objective

Create a clean consistent merchant-level dataset for downstream analysis



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ID Resolution

Updated merchant IDs using `LEFT JOIN + COALESCE`.

Added `is_mapped` flag for traceability.

Transaction Logic

Built running transaction counts utilizing:

`ROW_NUMBER()`

Cohort Dataset

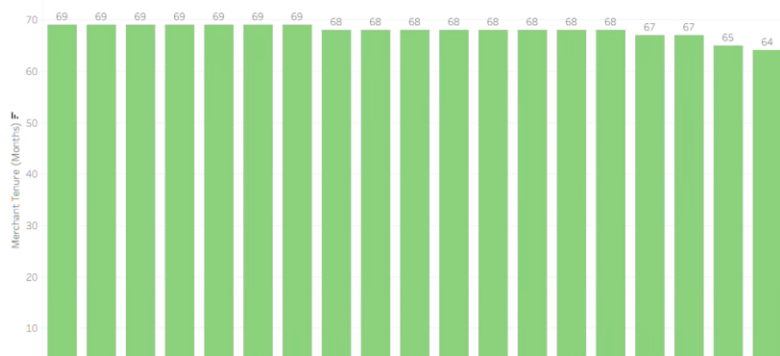
- Derived onboarding month from registration date
- Ranked merchants within cohorts using `RANK()`

Key Data Considerations

- Transaction counts per merchant are relatively consistent (10-22K per merchant)
- Merchant Onboarding is highly concentrated:
 - Majority of merchants onboarded in Aug-20, Sep-20, Dec-20, Apr-21

Tableau - Exploratory & Business Analysis

1. Merchant Tenure



- Measured time from first transaction to current date
- Provides view of merchant lifecycle distribution

2. MTD vs Previous MTD Value

MTD Performance vs Previous Month

(As of 03 Nov 2020)

Select Date

11/3/2020

Current MTD

Previous MTD

% Change

12.7M

16.5M

-23.1%

Current MTD is 23.1% lower than the same period last month, indicating a decline in transaction activity.

- Compared current vs prior month performance (same date range)

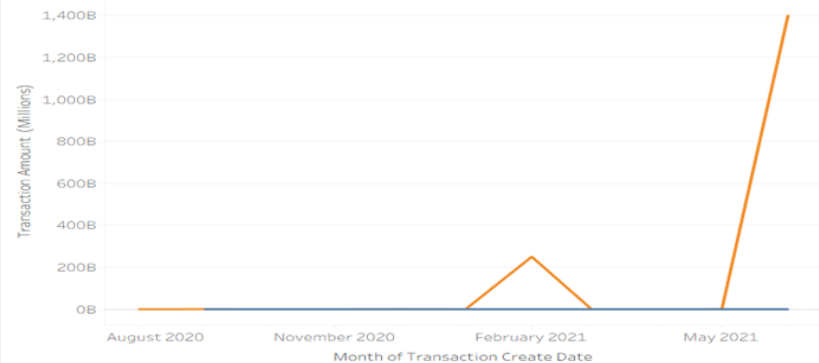
Key Observation: Aggregate trends are highly sensitive to top-merchant performance and tenure distribution.

Tableau - Exploratory & Business Analysis

3. Cohort based Transaction Trends

Cohort-Based Transaction Trends Over Time

(View: Detailed)

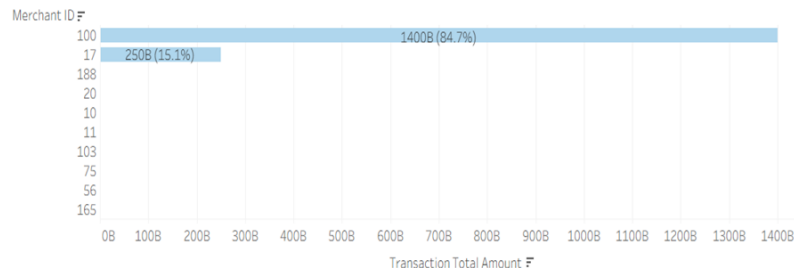


- Segmented merchants by onboarding month
- Tracked transaction behavior over time
- Toggle between cohort-level and overall view

4. Top 10 Merchants Analysis

Top 10 merchants ranked by total transaction value with percentage contribution

Merchant 100 contributes 84.7% and Merchant 17 contributes 15.1%, together accounting for ~99.8% of total TPV. The remaining 8 merchants contribute negligibly. 9 out of 10 top merchants are from HUN with one from GBR, indicating strong geographic concentration.



- Ranked merchants by transaction contribution as well as country, highlighting concentration within dataset
- Merchant 100 alone accounts for ~84.7% (~1.4T), while Merchant 17 contributes ~15.1% (~250B)

Key Observation: Aggregate trends are highly sensitive to top-merchant performance and tenure distribution.

Tableau Key Insights

Extreme Merchant Concentration

- Top 2 Merchants Contribute ~99% of total TPV.
- One merchant alone contributes ~85%.
- Majority of high value merchants concentrated in Hungary.
- *Indicates strong dependency on a small subset.*

Trends Driven by Outliers

- Large spikes in TPV by a few high-value transactions.
- MTD shows decline (~-23%) but heavily influenced by top merchants.
- *Aggregate trends are volatile and not representative.*

Cohort Insights are Limited

- Most merchants onboarded in a few months.
- Similar tenure across merchants (~63-68 Months).
- Aug 2020 cohort dominated activity over time.
- *Reduces meaningful differentiation across cohorts.*

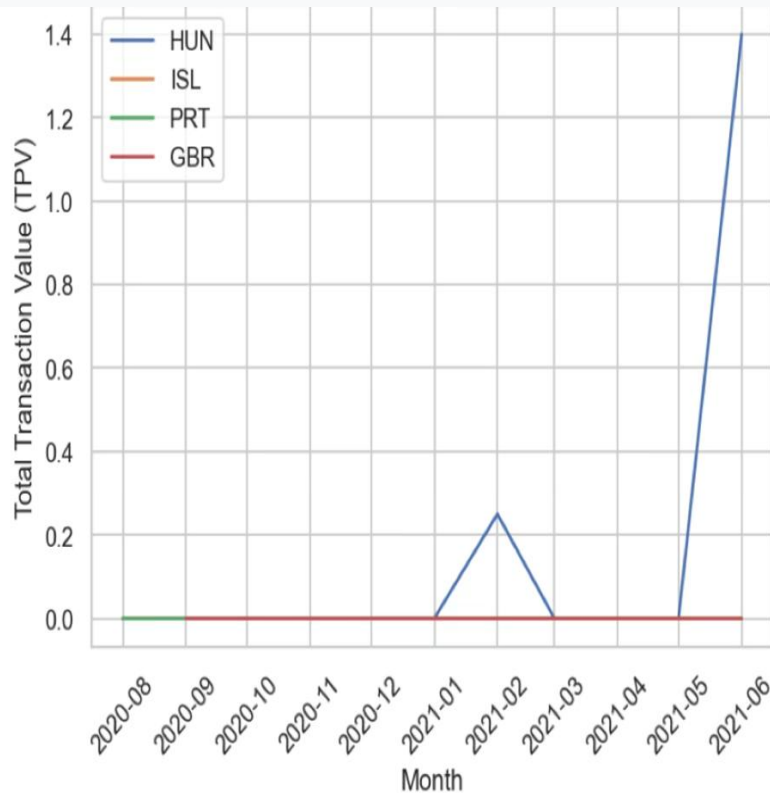
Summary: The dashboard reveals a merchant base highly sensitive to individual outliers, making aggregate trends volatile and cohort analysis challenging.

Python - Time Series & Behavioral Analysis

1. Geographic concentration

- Transaction value is highly concentrated in Hungary (HUN), which dominates TPV over time.
- Other countries contribute negligibly, indicating strong geographic concentration in transaction activity
- Transaction activity is concentrated in 4 countries - HUN, ISL, PRT, GBR

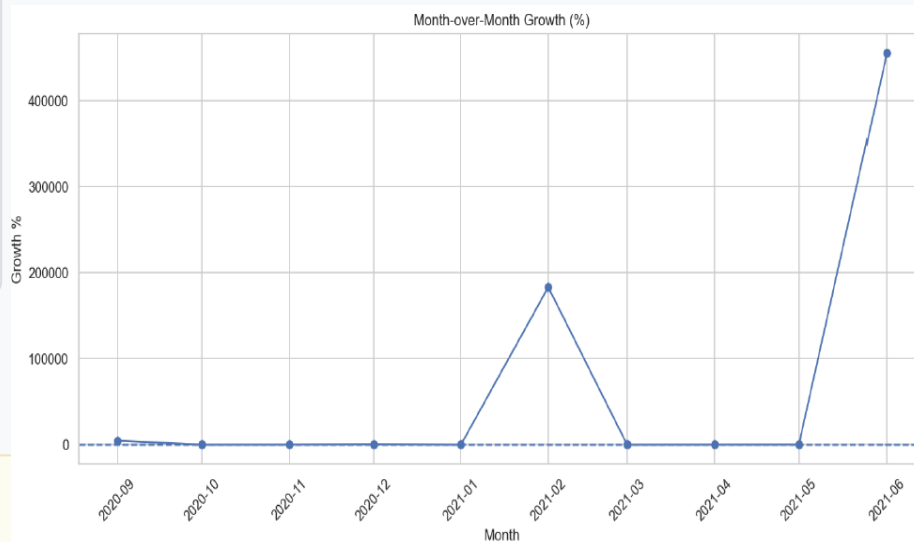
Key Behavioral Insight: HUN dominates transaction activity



Python - Time Series & Behavioral Analysis

2. No Clear Monthly Seasonality

- TPV remained flat from Aug–Dec 2020, with irregular spikes in Feb & Jun 2021
- Extreme MoM fluctuations driven by high-value transactions
- Growth trends are not stable or representative



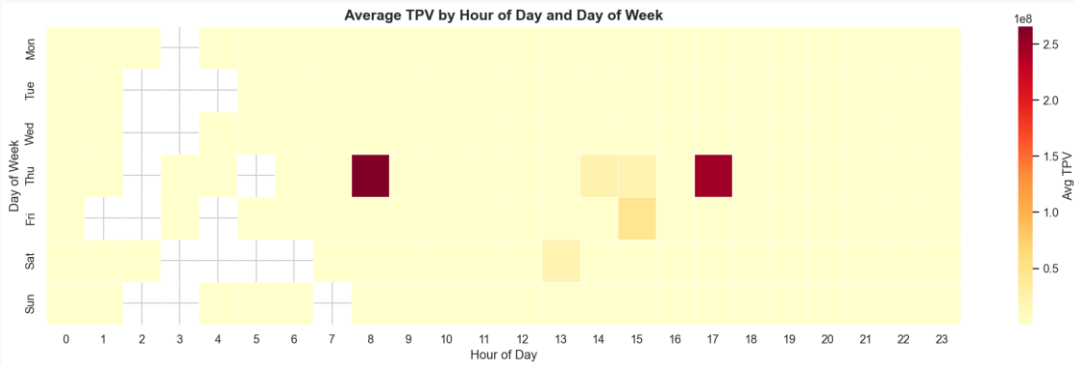
Key Behavioral Insight:

The MoM growth chart reflects the same spikes seen in monthly TPV. June 2021 shows ~455,000% growth which is not representative of organic growth. It is almost entirely attributable to a single merchant's transaction volume that month

Python - Time Series & Behavioral Analysis

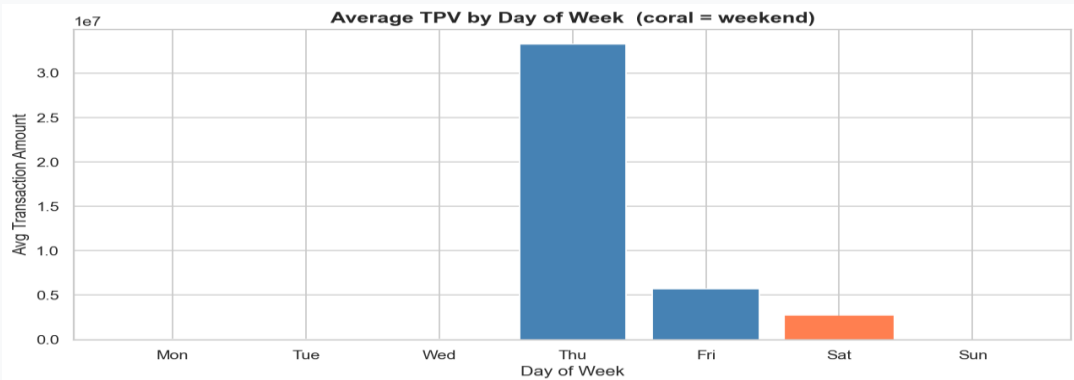
3. Hour x Day of Week

- Concentrated activity during business hours (7am–6pm)
- Distinct peaks on Thursdays 8 AM and 5 PM
- Post 6PM activity near zero
- Consistent with B2B payment behavior



4. TPV by Day of Week

- Thursday dominates TPV significantly, with Friday second
- Mon - Wed effectively zero
- Weekends are sparse with minimal activity



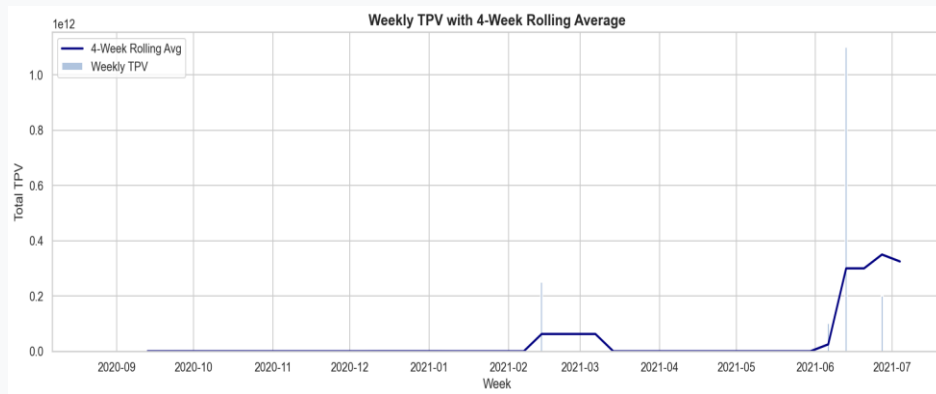
Key Observation:

Behavioral data confirms B2B-centric usage with peak activity on Thursdays and Fridays during business hours

Python - Time Series & Behavioral Analysis

5. Weekly TPV with Rolling Average

- Rolling average smooths out the two spike events (Feb and June 2021) and shows that underlying TPV was essentially flat until mid-2021.



Key Behavioral Insight:

The upward trend in the rolling average from June 2021 onward could indicate the start of real growth, though the dataset ends shortly after so this cannot be confirmed.

Key Takeaways & Next Steps

1. Key Takeaways

- High concentration in a small number of merchants
- Trends driven by high-value transactions
- No clear seasonality; irregular patterns
- Activity suggests business-driven behavior

2. Implications

- Aggregate metrics (TPV) can be misleading
- Strong dependency on limited merchants
- Trends don't reflect typical behavior

3. Next Steps

- Analyze top merchants separately
- Isolate outliers in trend analysis
- Focus on merchant-level performance
- Validate with balanced datasets

Conclusion: Data is heavily skewed and representative of 2 merchants largely; proper segmentation is critical to avoid misleading aggregate trends.